



Make OnePot Protein Mix

Getting started

You want to make PURE. You're going to have to make Protein Mix, which contains the thirty-six (36) proteins in the PURE system. If you want to purify each protein individually, you can follow our "36-pot" protocol ([👉 Make Protein Mix](#)). However, if you want working protein mix with less time and effort, you can co-express and co-purify all 36 proteins at once in "OnePot". This protocol shows you how. This protocol has been established to grow, purify, and characterize OnePot PURE proteins following Lavickova's OnePot PURE protocol [[Grasemann, 2021](#)] with a few deviations.

Materials and Equipment

▼ Table 1: Reagents and Consumables

Name	Product	Manufacturer	Part #	Price	Storage Conditions	Link
Columns						
Gravity columns	Empty Disposable PD-10 Columns	Cytiva	17043501	\$258.30	4C to 30C	[link]
Ni ²⁺ Affinity Resin	NEBExpress Ni Resin	New England Biolabs	S1428S	\$358.00	1C to 4C	[link]
Salts (for Buffers)						
Potassium Hydroxide	Potassium hydroxide - ACS reagent, ≥85%, pellets	Sigma-Aldrich	221473-1KG	\$99.90	4C to 30C	[link]
Hydrochloric acid (1M)	Hydrochloric acid 1 N, Reagent Grade, Packaging=Amber Bottle, Size=500 mL	VWR	E447-500L	\$72.46	4C to 30C	[link]
HEPES	HEPES, Crystalline Powder, ≥99.5% (titration), Poly bottle	Sigma-Aldrich	H3375-500G	\$431	4C to 30C	[link]
Imidazole	ReagentPlus® Imidazole, Crystalline, 99%	Sigma-Aldrich	I202-500G	\$120	4C to 30C	[link]
Ammonium Chloride	Ammonium chloride, ACS reagent, ≥99.5%	Sigma-Aldrich	213330-500G	\$73.50	4C to 30C	[link]
Sodium Chloride	Sodium Chloride, Redi-Dri™, anhydrous, free-flowing, ACS, ≥99%	Sigma-Aldrich	746398-1KG	\$133.00	4C to 30C	[link]
Potassium Chloride	Potassium Chloride, ACS reagent, 99.0-100.5%	Sigma-Aldrich	P3911-1KG	\$146.00	4C to 30C	[link]

Name	Product	Manufacturer	Part #	Price	Storage Conditions	Link
Magnesium Chloride hexahydrate	BioXtra Magnesium chloride, Hexahydrate, Powder or Crystals, ≥99.0%	Sigma-Aldrich	M2670-100G	\$50.50	4C to 30C	[link]
Supplements (for Buffers)						
Glycerol	Glycerol, Molecular Biology Grade, Liquid, ≥99.0%	Sigma-Aldrich	G5516-1L	\$157.00	4C to 30C	[link]
Lysozyme	Lysozyme from chicken egg white, protein ≥90%, ≥40,000 units/mg protein, lyophilized powder	Sigma-Aldrich	L6876-1G	\$76.90	-25C to -15C	[link]
TCEP-HCl (0.5M)	Thermo Scientific, Bond-Breaker® Tris[2-Carboxyethyl]phosphine Neutral Solution (TCEP), Application=Water-Soluble Reducing Agent, Molecular Weight=286.65, Size=5 mL	Thermo Scientific	77720	\$175.65	4C to 30C	[link]
cOmplete Protease Inhibitor	cOmplete™ Protease Inhibitor Cocktail, EDTA-Free, Tablets	Roche	11873580001	\$472	1C to 4C	[link]
Media						
LB	Luria Broth (Miller's LB Broth), Non-Sterile, 6.8 - 7.2, Molecular Biology Grade, suitable for regular E.coli culture, Powder	Sigma-Aldrich	L3522-1KG	\$221	4C to 30C	[link]
IPTG	Isopropyl β-D-thiogalactoside (IPTG), Powder, ≥99% (TLC), ≤0.1% Dioxane	Sigma-Aldrich	I6758-1G	\$89.90	-25C to -15C	[link]
Kanamycin	BioReagent Kanamycin sulfate, ≥750 ug/mg, From Streptomyces kanamyceticus, Suitable for cell culture, Suitable for plant cell culture, Powder	Sigma-Aldrich	K1377-1G	\$47.70	4C to 30C	[link]
Consumables						
Culture tubes	Culture Tube, PS, 14mL, 18×95mm, Sterile, TC Treated, w/ Snap (Vent) Cap	Greiner Bio-One	191180	\$258.15	4C to 30C	[link]

Name	Product	Manufacturer	Part #	Price	Storage Conditions	Link
15 mL centrifuge tubes	Corning® 15 mL PP Centrifuge Tubes, Rack Packed with CentriStar™ Cap, Sterile, 50/Rack, 500/Case	Corning	430790	\$230.44	4C to 30C	[link]
10 mL Luer lock syringes	Syringes with BD Luer-Lok® Tip, BD Medical, Syringe with Luer-Lok® Tip, Volume=10 mL	BD Industrial/Difco	550-80620-PK	\$102.40	4C to 30C	[link]
0.22 µm syringe filters	PES Syringe Filter, 0.22µm, 30mm, Sterile	CELLTREAT Scientific Products	229747	\$84	4C to 30C	[link]
0.22 µm vacuum filters	Corning® 500 mL Vacuum Filter/Storage Bottle System, 0.22 µm Pore 33.2cm² PES Membrane, Sterile, 12/Case	Corning	431097	\$187.14	4C to 30C	[link]
0.45 µm syringe filters	PES Syringe Filter, 0.45µm, 30mm, Sterile	CELLTREAT Scientific Products	229749	\$84	4C to 30C	[link]
3 kDa, 15 mL centrifugal filters	Amicon® Ultra Centrifugal Filter, 3 kDa MWCO	Millipore Sigma	UFC9003	\$125	4C to 30C	[link]
3 kDa, 0.5 mL centrifugal filters	Amicon® Ultra Centrifugal Filter, 3 kDa MWCO	Millipore Sigma	UFC5003	\$63	4C to 30C	[link]
Tris-Glycine 4% - 20% gels	4–20% Mini-PROTEAN® TGX™ Precast Protein Gels, 10-well, 50 µl	BioRad	4561094	\$153.00	1C to 4C	[link]
96-well optical plate	Microplate, 96 well, PS, U-bottom, clear	Greiner	650101	\$156.14	4C to 30C	[link]
Glassware						
2L baffled flasks	PYREX® 2L Delong Shaker Erlenmeyer Flask with Extra Deep Baffles	Pyrex	4446-2L	\$233.98	4C to 30C	[link]
flask closures	Chemglass Life Sciences Closure, 38mm, Stainless Steel	Chemglass Life Sciences	Chemglass Life Sciences Closure, 38mm, Stainless Steel	\$100.75	4C to 30C	[link]

▼ Table 2: Equipment

Name	Product	Manufacturer	Part #	Price	Link
Thermocycler	SimpliAmp™ Thermal Cycler	Applied Biosystems	A24811	\$6,580.00	[link]
Shaking incubator	New Brunswick Innova 4430 Incubator Shaker	New Brunswick	-	-	discontinued
Floor centrifuge	Sorvall LYNX 6000 Superspeed Centrifuge	Sorvall	75006590	unlisted	[link]
Sonicator	Q700 Sonicator	QSonica	Q700A-110	\$5500	[link]

Name	Product	Manufacturer	Part #	Price	Link
Sonicator Probe	Standard Probe, Replaceable Tip, 1/2" (12.7mm) diameter	QSonica	4220	\$670	[link]
Sound Enclosure	Sound Enclosure	QSonica	432B2	\$995	[link]
Lab Jack	Fisherbrand™ Lab Jacks (15cmx15cm)	Fisher Scientific	14-673-51	\$484	[link]
Thermocycler	SimpliAmp™ Thermal Cycler	Applied Biosystems	A24811	\$6,580.00	[link]
Gel Tank	Criterion™ Cell	BioRad	1656001	\$852.00	[link]
Gel Power Supply	PowerPac™ Basic Power Supply	BioRad	1645050	\$530.00	[link]
Gel Imager	ChemiDoc™ Imaging System	BioRad	12003153	unlisted	[link]
-20C Freezer	TSX Series High-Performance -20°C Manual Defrost Freezers	Thermo Scientific	TSX2320FA	unlisted	[link]
-80C Freezer	TSX Series Ultra-Low Freezers	Thermo Scientific	TSX60086A	unlisted	[link]

Protocol - Day 0

☐ Buffer Preparation

- ☐ Please use the [Nucleus Buffer Template](#). Instructions on how to use template below:

▼ Template Usage Note:

Modify highlighted values in blue. Check units carefully. Output values shown in purple.

☐ Prepare Stock Solutions:

- ☐ Fill water to 75% of total volume.
- ☐ Weigh and dissolve required mass of salt (check the listed molecular weight of your salts and update if necessary; hydration salts can have different molecular weights)
- ☐ Adjust pH if specified. Use KOH (10M) or HCl (1M)
- ☐ Adjust water to final volume.

☐ Prepare Buffer Solutions:

- ☐ Add required volume of each stock solution and adjust to final volume with water.
- ☐ Withhold TCEP until day of use.

▼ Notes: Protein Storage Buffers are prepared from concentrate (4x)

You will prepare a concentrated (4x) protein storage stock. This buffer can be diluted to 1x with varying amount of glycerol and water to make Dialysis Buffer (0% glycerol), 30% glycerol Protein Storage Buffer, and 60% glycerol Protein Storage Buffer.

Protocol - Day 1

☐ Prepare starter cultures (night before)

- ☐ Working under flame, Prepare the media by mixing 20 mL LB with 20 uL of Kan (50 mg/mL).
- ☐ In a deep-well plate, add 300 uL of media into 35 wells and inoculate each well with the 35 PURE proteins (omitting EF-Tu). Seal the plate with a breathable membrane and incubate at 37C / 260 rpm / overnight.
- ☐ For EF-Tu, inoculate 3 mL of LB + Kan media in a 14 mL cell culture tube and incubate at 37C/ 260 rpm/ (12 - 16) hrs.

☐ Inoculate Co-culture

☐ (Optional) Confirm all strains have grown by measuring the OD600 on a 96-well plate with 10x dilutions or by visually confirming turbidity.

☐ Transfer 500 mL LB + Kan into baffled 2L Erlenmeyer flask and inoculate with the starter cultures:

- ☐ 5 mL total inoculation volume (~140x back-dilution).
- ☐ 2.33 mL of EF-Tu culture (~300x back-dilution).
- ☐ 75 μ L of each culture from deep-well plate into reagent reservoir.

☐ **Outgrowth & Harvest**

- ☐ Thaw IPTG at 4C.
- ☐ Incubate culture at 37C/ 260 rpm/ ~2.5 hours or until the OD600 reaches 0.6-0.8.
 - ☐ At +2 hours, measure OD600 from 1 mL sample in a cuvette. Repeat every 20 minutes until OD600 reaches target.
 - ☐ Mark inoculation time.
- ☐ Induce protein expression with 500 μ L of 0.5 M IPTG and incubate at 37C / 260 rpm / 2.5 hours. Mark induction time.
- ☐ Harvest the cells via centrifugation at 16,000 x g / 4C / 15 minutes.
 - ☐ Tare plastic bag on a coarse scale.
 - ☐ Decant supernatant into biohazardous waste.
 - ☐ Scoop cell pellet ("cellet") into pre weighed ziploc bag. Weigh and record mass on bag.
 - ☐ (Optional) flash freeze plastic bag in liquid nitrogen.

☐ **Storage**

- ☐ Store the cellet at -80C.

Protocol - Day 2

☐ **Finish Buffers**

- ☐ Add TCEP (0.5M, 500x) to finish Lysis Buffer, Wash Buffer, Elution Buffer, Dialysis Buffer, and Storage Supplement Buffer. E.g., 50 mL Wash Buffer requires 100 μ L TCEP.
- ☐ Resuspend one protease inhibitor tablet per four protein preps in 4 mL Lysis Buffer in a 15 mL conical tube. Resuspend by vortexing vigorously. Add one quarter of the resuspended protease inhibitor mixture (1 mL) to your tube of Lysis Buffer and mix by inversion or vortexing.

☐ **Cell lysis**

- ☐ Resuspend cellet in its plastic bag with 10 mL Protein Wash Buffer. Massage cellet until suspended.
- ☐ Transfer resuspended to 15 mL conical tube and keep on ice.
- ☐ Lyse cells by sonication using a 130-watt probe sonicator (6 mm diameter probe). Set sonicator to 25% amplitude/ 15 seconds on/30 seconds off. Sonicate samples on ice for 2 minutes on-time. It may take 2-4 cycles to sufficiently lyse cells. Do NOT let the sample heat up. If so, allow the sample to cool on ice before continuing sonication.
- ☐ Clarify lysate by centrifugation at 16,000 x g/ 4C/ 20 min.
- ☐ Use a 10 mL syringe with 0.22 μ m syringe filter to filter the supernatant into a fresh 15 mL tube.
- ☐ Discard pellet in biohazardous waste.

▼ **Note: Visually inspect pellet for lysis**

White material in pellet indicates lysed cell mass. A sufficiently lysed pellet should appear smaller than the original pellet, lighter in color, and have a translucent ring around the pellet. Note if you see black material at the base of the pellet. This indicates oversonication.

☐ **Column Prep**

☐ Assemble columns.

- ☐ Cut tips off of columns off using a razor blade.
- ☐ Pack a filter into the bottom of the column (we use the back end of a cell spreader).

▼ **Notes: columns have ridges!**

The filter needs to be pushed across two circular ridges on the inside of the column, one ~2 cm from the opening of the column and another ~5 cm further down the column

- ☐ Assemble buffer reservoir to column.

☐ Wash empty columns.

- ☐ Put column assembly on your column holder over your flowthrough catch.
- ☐ Wash column, buffer reservoir, and filter with ≥ 10 CV (20 mL) Ultrapure water. Discard flowthrough.

▼ **Notes: "CV" = Column Volume**

One "Column Volume" is equal to the volume of affinity resin in your column. For example, "wash with 5 CV of Wash Buffer" means that for 2 mL of resin, wash with 10 mL of Wash Buffer. Specifying buffer volumes by CVs allows practitioners to increase or decrease the scale of the purification and still use the same protocol.

☐ Load and Equilibrate Ni²⁺ affinity Resin into columns.

- ☐ Resuspend Ni²⁺ resin (50% v/v suspension) by shaking.
- ☐ Add 2 mL of resuspended resin by pipette to the column.
- ☐ Wash column with ≥ 10 CV (20 mL) mL Ultrapure water. Discard flowthrough.
- ☐ Equilibrate columns with ≥ 10 CV (20 mL) of Wash Buffer (4C). Discard flowthrough.

☐ **(Optional)** store Columns.

- ☐ Seal each column with a cap and bung.
- ☐ Store columns at 4C for up to 48 hrs.

▼ **Notes: store columns long term in EtOH (20%)**

Packed and loaded columns can harbor bacterial growth, even at 4C. If you are not going to use your columns within 48 hours of preparation, or if you want to save your used columns for later reuse, store them in EtOH (20%). Wash the columns as described above in Wash Buffer, then wash with 10 CV Ultrapure water to remove salts. Fill the gravity column with EtOH (20%), seal bung and lid with parafilm, and store at 4C until ready for use.

☐ **Gravity Column Ni-resin Purification**

- ☐ Add bung to gravity column to prevent flowthrough.
- ☐ Add clarified lysate (supernatant) to the equilibrated column.
- ☐ Seal top of column with cap.
- ☐ Incubate at 4C / 1hr with rotation.
- ☐ Mount column on a stand with a beaker below to capture flow through.
- ☐ Remove bung and let the flow through elute off the column.
- ☐ Wash column with 12.5 CV (25 mL) of Wash buffer and discard flow though.
- ☐ Elute sample with 2.5 CV (5 mL) Elution Buffer and capture eluent in 15 mL tube. This is your sample.

☐ **Buffer Exchange by diafiltration**

- ☐ Dilute your eluent (5 mL) with 10 mL Dialysis Buffer (3x dilution) and transfer to a 15 mL centrifugal filter.
- ☐ Spin samples at 4000 g/ 4C / 1 hr. Check your samples frequently (~20 min) and note the volume. Target $\leq$ 500 μ L.
- ☐ Dilute your sample with 15 mL fresh Dialysis Buffer ($\geq$ 90x dilution) and spin down again to $\leq$ 500 μ L.
- ☐ (Optional) dilute and spin your samples again with 15 mL Dialysis Buffer ($\geq$ 270x dilution).

☐ **Concentrate buffer exchanges proteins**

- ☐ Pipette mix the sample in the spin filter to resuspend any proteins that have collected on the filter.
- ☐ Check the concentration of your Protein Mix by A280. Calculate your estimated final volume to target (25 - 30) μ g / μ L.

▼ **Note: A280 overestimates protein concentration**

PURE uses Protein Mix at 20 μ g / μ L. We will use Pierce660, a robust and accurate colorimetric assay, to do precise quantitation of your protein concentration at a later step. Here, we just want a quick estimate of the concentration at this step.

A fast A280 measurement can be made with a microvolume spectrophotometer, however the resulting estimate is biased. It is easier to dilute samples than concentrate them, therefore we will aim to overconcentrate our proteins today and dilute them later.

- ☐ Transfer your samples to a smaller centrifugal filter (4 mL or 0.5 mL, depending on your target concentration) and spin to your target concentration.
- ☐ Spin your sample down to ~20 μ g / μ L.
- ☐ Measure your sample volume by reverse pipetting.
- ☐ Add an equal volume of Storage Supplement Buffer (60% glycerol) and spin until you reach your target concentration. Verify your concentration by A280.

▼ **Note: Proteins “crash out” at high concentrations**

Concentrating protein samples above 30 μ g / μ L can cause proteins to aggregate and precipitate out of solution. To make protein stocks in 30% glycerol at 20 μ g / μ L, you would need to concentrate them first to 60 μ g / μ L, then add an equal volume of Storage Supplement. This is impractical.

As a result, we add Storage Supplement before reaching our target concentration. However, high viscosity buffers (e.g., 30% glycerol) greatly slow down centrifugal filtration. Thus, we add storage supplement *just before* proteins crash out, and finish concentrating in their final buffer.

☐ **Storage**

- ☐ Transfer proteins to Protein low bind microfuge tubes.
- ☐ Store at -80C.

☐ **Quality Control**

- ☐ Visualize purity by 🍷 [Protein Gel](#)
- ☐ Determine concentration by 🌈 [Pierce660 Assay](#)

Resources and References

• Papers

- Lavickova, B. & Maerkl, S. J. A Simple, Robust, and Low-Cost Method To Produce the PURE Cell-Free System. ACS Synth. Biol. 8, 455–462 (2019). [\[link\]](#)
- 1. Grasemann, L., Lavickova, B., Elizondo-Cantú, M. C. & Maerkl, S. J. OnePot PURE Cell-Free System. JoVE (2021) doi:10.3791/62625. [\[link\]](#)

Credits

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